Rover Evidence Lab

GROW · Science · 40 minutes

I can carry out a comparative fair test, use repeated distance results for a conclusion and improve the method.

Prepare before pupils arrive

Setup: Prepare the W13A smooth-versus-fabric setup or the fixed new-investigation data. Mark a clear lane and print one six-row results sheet per pupil route.

Safety: Use a low stable ramp, one release point and a catch area. Stop if the rover leaves the lane; preserve the note, repair and repeat.

Equipment: Stable low ramp; same rover; smooth and fabric surface conditions; ruler or tape; six-row table; lane markers.

Safety: Teacher checks the ramp and clear lane. Release without pushing; keep hands, feet and faces away from the moving rover.

Fallback: Direct an adult through the physical method or use printed fixed data. Observation and decision roles are complete participation.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Consolidate and revisit key ideas through investigation.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	State the surface-test independent and dependent variables. Name two controls from the plan.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 4 min	repeat — same method again conclusion — what the results support limitation — what the test cannot show Check the safe lane, low ramp, marked start and same rover. Commit a prediction before releasing anything.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	New investigation evidence: smooth 48, 51, 49 cm; fabric 27, 29, 28 cm. Every smooth result is greater.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each evidence card under Smooth, Fabric or Conclusion. Freeze the pattern before choosing an improvement.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Carry out or direct the fair rover test, record two conditions with three repeats, then give a conclusion, improvement and limitation. Record: Prediction, all six raw results, pattern statement, evidence-linked conclusion, one improvement and one model limitation.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Use one repeated result pattern to explain the effect of surface on the rover. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Record all three valid results for each condition using the same measuring rule.

Yes. All valid repeats show the pattern and the variation without selecting only preferred results.

Misconception repair

Repaired. Preserve all valid raw results; a summary can be added but must not replace the evidence.

Guided checkpoint

Yes. All valid repeats show the pattern and the variation without selecting only preferred results. Choose the method that preserves every valid observation and uses the same rule.

48, 51, 49 cm

SMOOTH. These are all the smooth-surface results. Their typical value is about 49 cm. Match the raw values to the condition where they were measured.

Range: 3 cm (48–51 cm interval)

SMOOTH. The smooth numerical range is 3 cm; the observed values spread from 48 to 51 cm. For the smooth repeats, subtract the minimum 48 from the maximum 51.

27, 29, 28 cm

FABRIC. These are all the fabric-surface results. Their typical value is 28 cm. Place the smaller repeated distances with the fabric condition.

Range: 2 cm (27–29 cm interval)

FABRIC. The fabric numerical range is 2 cm; the observed values spread from 27 to 29 cm. For the fabric repeats, subtract the minimum 27 from the maximum 29.

Smooth produced greater travel distance

CONCLUSION / LIMIT. This conclusion matches all six results and explains the pattern using friction. A conclusion combines the comparison with the scientific reason.

One classroom rover is a simplified model

CONCLUSION / LIMIT. This honest limitation keeps the conclusion within the conditions actually tested. A limit belongs beside the conclusion because it states where the evidence may not transfer.

Model limitation

A short classroom ramp tests one rover, two surfaces and one set of conditions; it cannot represent gravity, terrain or every real rover design.

Independent evidence

Prediction, all six raw results, pattern statement, evidence-linked conclusion, one improvement and one model limitation.

Supported: Direct each run, match the six values to conditions and complete an oral or symbol-supported conclusion. Adult handles equipment; use pre-labelled rows and the frame: On ___ it went ___ because ___.

Standard: Complete three safe repeats per condition, record all results and write a conclusion plus one improvement. Check the method against change, measure, keep and repeat before concluding.

Stretch: Compare typical values, calculate numerical ranges, explain variation and evaluate transfer to a real rover. Suggest one justified method improvement without changing the original raw data.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W13B_Rover_Rescue_Investigation_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.